

Due Date: 11:59 pm on 2/17/2026

Instructor: Dr. Tutku Karacolak

## ECE 370 Electromagnetic Fields and Waves

### HOMEWORK #3

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| Problem | Course outcome | Grade |
|---------|----------------|-------|
| 1       | 1-c, 1-d       | /30   |
| 2       | 1-a, 1-d       | /30   |
| 3       | 1-c, 1-d       | /40   |
| TOTAL:  |                | /100  |

1- Solve Problem 4.32 from the textbook.

**4.32** A circular ring of charge of radius  $a$  lies in the  $x$ - $y$  plane and is centered at the origin.

Assume also that the ring is in air and carries a uniform density  $\rho_\ell$ .

**a.** Show that the electrical potential at  $(0, 0, z)$  is given by

$$V = \rho_\ell a / [2\epsilon_0 (a^2 + z^2)^{1/2}].$$

**b.** Find the corresponding electric field  $\mathbf{E}$ .

2- A ceramic sphere of radius  $\frac{1}{2}$  m is charged with a surface charge density  $\rho_s = 1.5$  C/m<sup>2</sup>. The relative permittivity of the ceramic is  $\epsilon_r = 4$ . The region outside of the sphere has  $\epsilon_r = 2$ . The electric flux density on the outside surface of the ceramic is

$$\vec{D} = 0.7\hat{a}_R + 2\hat{a}_\theta - 3\hat{a}_\phi \quad (\text{C/m}^2)$$

Use electric field boundary conditions to find the electric flux density on the inside surface of the sphere.

3- The potential in free space is given by:

$$V(x, y, z) = -x^2y(z^2 + 1) \text{ (V)}$$

Determine

- (a) the vector electric field. (12 pts)
- (b) the vector electric flux density. (8 pts)
- (c) the potential difference  $V_{AB}$  between  $A=(1, 2, 5)$  m and  $B=(1, 2, 1)$  m by integrating the electric field along the straight line path between these two points. (20 pts)